

GraphTheorySymbol: L^AT_EX Commands with Symbols for Graph Theory

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Abstract

This documentation is for L^AT_EX package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

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1 Introduction

This documentation is for $\text{\LaTeX}$ package `GraphTheorySymbol`. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com/LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\circ$	$u \circ v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidence/</code>	<code>\GTSoI/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSoIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\circ$	$v \circ e$
Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\circ$	$u \circ v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\circ$	$u \circ v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\circ$	$u \circ v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\circ$	$u \circ v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\circ$	$u \circ v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\circ$	$v \circ e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSoIM/</code>	$\circ$	$v \circ e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSoIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\circ$	$v \circ e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\circ$	$v \circ e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\circ$	$v \circ e$

\GTSsetMainColor{blue}			
Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		$u \overset{\circ}{\circ} v$
\GTSadjacencyRelation/	\GTSaR/		$u \overset{\circ}{\circ} v$
\GTSunadjacency/	\GTSu/		$u \overset{\circ}{\circ} v$
\GTSunadjacencyRelation/	\GTSuR/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyUp/	\GTSdAU/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyDown/	\GTSdAD/		$u \overset{\circ}{\circ} v$
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		$u \overset{\circ}{\circ} v$
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		$u \overset{\circ}{\circ} v$
\GTSincidence/	\GTSi/		$v \overset{\circ}{\circ} e$
\GTSincidenceRelation/	\GTSiR/		$v \overset{\circ}{\circ} e$
\GTSnegatedIncidence/	\GTSnI/		$v \overset{\circ}{\circ} e$
\GTSnegatedIncidenceRelation/	\GTSnIR/		$v \overset{\circ}{\circ} e$
\GTSoutIncidence/	\GTSOI/		$v \overset{\circ}{\circ} e$
\GTSoutIncidenceRelation/	\GTSOIR/		$v \overset{\circ}{\circ} e$
\GTSnegatedOutIncidence/	\GTSnOI/		$v \overset{\circ}{\circ} e$
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		$v \overset{\circ}{\circ} e$
\GTSinIncidence/	\GTSiI/		$v \overset{\circ}{\circ} e$
\GTSinIncidenceRelation/	\GTSiIR/		$v \overset{\circ}{\circ} e$
\GTSnegatedInIncidence/	\GTSnII/		$v \overset{\circ}{\circ} e$
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		$v \overset{\circ}{\circ} e$

Long name	Short name	Result	In context
\GTSmetaNot/	\GTSmN/	$\neg$	$\neg \overset{\circ}{\circ}$
\GTSmetaAnd/	\GTSmA/	$\wedge$	$\overset{\circ}{\circ} \wedge \overset{\circ}{\circ}$
\GTSmetaOr/	\GTSmO/	$\vee$	$\overset{\circ}{\circ} \vee \overset{\circ}{\circ}$

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Directed graph:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Oriented graph:

$\forall u, v, u \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} v$.

Tournament adjacency:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with $\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\}$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with $\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\}$.

Directed graph:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with

$\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSdAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSnDAU}/\}$.

Oriented graph:

$\overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ} \overset{\circ}{\circ}$ is obtained with

$\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSnDAU}/\}$.

Prefix notation:

$\exists u, v, u \left[\overset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\text{V}}}}} \right] v \Leftrightarrow \exists u, v, u \left[\overset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\text{V}}}}} \right] v.$

$\exists v, e, v \left[\overset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\text{V}}}}} \right] e \Leftrightarrow \exists v, e, v \left[\overset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\text{V}}}}} \right] e.$

$\left[\overset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\text{V}}}}} \right]$ is obtained with

`\mathrel{\{[\backslashGTSmO/\backslashGTSxa[mC=red,d]\backslashGTSxa[mC=green,d]\backslashGTSxa[mC=blue,u]]\}}.`

$\left[\overset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\underset{\text{m}}{\text{V}}}}} \right]$ is obtained with

`\mathrel{\{[\backslashGTSmO/\backslashGTSxi[mC=red,i]\backslashGTSxi[mC=green,i]\backslashGTSxi[mC=blue,o]]\}}.`

Apart from these symbol macros, versatile macros to call them all are available with options:

`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,

`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,

`\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols,

`\GTSx[]{}` for all symbols

(`\GTSx[]{}a` = `\GTSxa[]`, `\GTSx[]{}i` = `\GTSxi[]`, `\GTSx[]{}m` = `\GTSxml[]`).

See Detailed usage section for full syntax of these macros.

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

For completeness, we also documented macros with an at sign, but as usual, you should not (have to) use them directly.

2.2.1 General macros

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{<commandname>}` The name of the command to create.

`{<commandcode>}` The code of the command to create.

See its [implementation](#) on page 26.

`\GTS@newprotectedcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{<commandname>}` The name of the command to create.

`{<commandcode>}` The code of the command to create.

See its [implementation](#) on page 27.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{<color>}` The color that must be used by GraphTheorySymbol afterward.

See its [implementation](#) on page 27.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{<color>}` The color that must be used by GraphTheorySymbol afterward.

See its [implementation](#) on page 27.

2.2.2 Standard adjacency macros

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{<mainColor>}` If non-empty, this color will be used as main color.

`{<secondColor>}` If non-empty, this color will be used as second color.

`{<drawSlash>}` Non-empty implies that a negation will be drawn.

`{<drawUpArrow>}` Non-empty implies that an up-arrow will be drawn.

`{<drawDownArrow>}` Non-empty implies that a down-arrow will be drawn.

See its [implementation](#) on page 27.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

`{<mainColor>}` If non-empty, this color will be used as main color.

`{<secondColor>}` If non-empty, this color will be used as second color.

`{<drawSlash>}` Non-empty implies that a negation will be drawn.

$\{\langle drawUpArrow \rangle\}$ Non-empty implies that an up-arrow will be drawn.
 $\{\langle drawDownArrow \rangle\}$ Non-empty implies that a down-arrow will be drawn.
 $\{\langle relation \rangle\}$ Non-empty implies that relation spacing will be added.
See its [implementation](#) on page 28.

$\backslash\text{GTSxadjacency}$, $\backslash\text{GTSxadj}$, $\backslash\text{GTSxa}$ Versatile macros for adjacency that can be given options.

Argument:

$[\langle optionsList \rangle]$ Sets the following options if not empty:

$\text{mainColor}=\text{color1}$ or $\text{mainC}=\text{color1}$ or $\text{mColor}=\text{color1}$ or $\text{mC}=\text{color1}$,
 $\text{secondColor}=\text{color2}$ or $\text{secondC}=\text{color2}$ or $\text{sColor}=\text{color2}$ or $\text{sC}=\text{color2}$,
 monochrome or mono or mon or m (equivalent to $\text{mC}=\text{black}, \text{sC}=\text{black}$),
 negated or not or n ,
 up or u ,
 down or d ,
 relation or rel or r .

See their [implementations](#) on page 28.

$\backslash\text{GTSadjacency}$, $\backslash\text{GTSadj}$, $\backslash\text{GTSa}$ These macros will draw an adjacency symbol. A first drawing didn't include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as $\text{Adj}(\dots)$.

$\text{A}.\backslash\text{GTSadjacency}/\text{A}.$

or $\text{A}.\backslash\text{GTSadj}/\text{A}.$

or $\text{A}.\backslash\text{GTSa}/\text{A}.$

will yield $\text{A} \circ \text{A}.$

$\text{u} \backslash\text{GTSadjacency}/ \text{v}$

or $\text{u} \backslash\text{GTSadj}/ \text{v}$

or $\text{u} \backslash\text{GTSa}/ \text{v}$

will yield $\text{u} \circ \text{v}.$

$\$u \backslash\text{GTSadjacency}/ v\$$

or $\$u \backslash\text{GTSadj}/ v\$$

or $\$u \backslash\text{GTSa}/ v\$$

will yield $u \circ v.$

See their [implementations](#) on page 29.

$\backslash\text{GTSadjacencyRelation}$, $\backslash\text{GTSadjRel}$, $\backslash\text{GTSaR}$ These macros will draw an adjacency symbol with relation spacing in math mode.

$\$u \backslash\text{GTSadjacencyRelation}/ v\$$

or $\$u \backslash\text{GTSadjRel}/ v\$$

or $\$u \backslash\text{GTSaR}/ v\$$

will yield $u \circ v.$

See their [implementations](#) on page 29.

$\backslash\text{GTSunadjacency}$, $\backslash\text{GTSunadj}$, $\backslash\text{GTSu}$ These macros will draw an unadjacency symbol.

$\text{A}.\backslash\text{GTSunadjacency}/\text{A}.$

or $\text{A}.\backslash\text{GTSunadj}/\text{A}.$

or $\text{A}.\backslash\text{GTSu}/\text{A}.$

will yield $A \circ A$.

`u \GTSunadjacency/ v`

or `u \GTSunadj/ v`

or `u \GTSu/ v`

will yield $u \circ v$.

`$u \GTSunadjacency/ v$`

or `$u \GTSunadj/ v$`

or `$u \GTSu/ v$`

will yield $u \circ v$.

See their [implementations](#) on page 29.

\GTSunadjacencyRelation, \GTSunadjRel, \GTSuR These macros will draw an unadjacency symbol with relation spacing in math mode.

`$u \GTSunadjacencyRelation/ v$`

or `$u \GTSunadjRel/ v$`

or `$u \GTSuR/ v$`

will yield $u \circ v$.

See their [implementations](#) on page 29.

\GTSdirectedAdjacencyUp, \GTSdirAdjUp, \GTSdAU These macros will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

`A.\GTSdirectedAdjacencyUp/A.`

or `A.\GTSdirAdjUp/A.`

or `A.\GTSdAU/A.`

will yield $A \circ A$.

`u \GTSdirectedAdjacencyUp/ v`

or `u \GTSdirAdjUp/ v`

or `u \GTSdAU/ v`

will yield $u \circ v$.

`$u \GTSdirectedAdjacencyUp/ v$`

or `$u \GTSdirAdjUp/ v$`

or `$u \GTSdAU/ v$`

will yield $u \circ v$.

See their [implementations](#) on page 29.

\GTSdirectedAdjacencyUpRelation, \GTSdirAdjUpRel, \GTSdAUR These macros will draw a directed adjacency upward symbol with relation spacing in math mode.

`$u \GTSdirectedAdjacencyUpRelation/ v$`

or `$u \GTSdirAdjUpRel/ v$`

or `$u \GTSdAUR/ v$`

will yield $u \circ v$.

See their [implementations](#) on page 29.

`\GTSnegatedDirectedAdjacencyUp`, `\GTSnegDirAdjUp`, `\GTSnDAU` These macros will draw a negated directed adjacency upward symbol.

`A.\GTSnegatedDirectedAdjacencyUp/A.`

or `A.\GTSnegDirAdjUp/A.`

or `A.\GTSnDAU/A.`

will yield $A \overset{\circ}{\circ} A.$

`u \GTSnegatedDirectedAdjacencyUp/ v`

or `u \GTSnegDirAdjUp/ v`

or `u \GTSnDAU/ v`

will yield $u \overset{\circ}{\circ} v.$

`$u \GTSnegatedDirectedAdjacencyUp/ v$`

or `$u \GTSnegDirAdjUp/ v$`

or `$u \GTSnDAU/ v$`

will yield $u \overset{\circ}{\circ} v.$

See their [implementations](#) on page 29.

`\GTSnegatedDirectedAdjacencyUpRelation`, `\GTSnegDirAdjUpRel`, `\GTSnDAUR` These macros will draw a negated directed adjacency upward symbol with relation spacing in math mode.

`$u\GTSnegatedDirectedAdjacencyUpRelation/v$`

or `$u\GTSnegDirAdjUpRel/v$`

or `$u\GTSnDAUR/v$`

will yield $u \overset{\circ}{\circ} v.$

See their [implementations](#) on page 30.

`\GTSdirectedAdjacencyDown`, `\GTSdirAdjDown`, `\GTSdAD` These macros will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

`A.\GTSdirectedAdjacencyDown/A.`

or `A.\GTSdirAdjDown/A.`

or `A.\GTSdAD/A.`

will yield $A \underset{\circ}{\circ} A.$

`u \GTSdirectedAdjacencyDown/ v`

or `u \GTSdirAdjDown/ v`

or `u \GTSdAD/ v`

will yield $u \underset{\circ}{\circ} v.$

`$u \GTSdirectedAdjacencyDown/ v$`

or `$u \GTSdirAdjDown/ v$`

or `$u \GTSdAD/ v$`

will yield $u \underset{\circ}{\circ} v.$

See their [implementations](#) on page 30.

`\GTSdirectedAdjacencyDownRelation`, `\GTSdirAdjDownRel`, `\GTSdADR` These macros will draw a directed adjacency downward symbol with relation spacing in math mode.

`\$u\GTSdirectedAdjacencyDownRelation/v\$`

or `\$u\GTSdirAdjDownRel/v\$`

or `\$u\GTSdADR/v\$`

will yield $u \Downarrow v$.

See their [implementations](#) on page 30.

`\GTSnegatedDirectedAdjacencyDown`, `\GTSnegDirAdjDown`, `\GTSnDAD` These macros will draw a negated directed adjacency downward symbol.

`A.\GTSnegatedDirectedAdjacencyDown/A.`

or `A.\GTSnegDirAdjDown/A.`

or `A.\GTSnDAD/A.`

will yield $A.\nexists A.$

`u \GTSnegatedDirectedAdjacencyDown/ v`

or `u \GTSnegDirAdjDown/ v`

or `u \GTSnDAD/ v`

will yield $u \nexists v$.

`\$u \GTSnegatedDirectedAdjacencyDown/ v\$`

or `\$u \GTSnegDirAdjDown/ v\$`

or `\$u \GTSnDAD/ v\$`

will yield $u \nexists v$.

See their [implementations](#) on page 30.

`\GTSnegatedDirectedAdjacencyDownRelation`, `\GTSnegDirAdjDownRel`, `\GTSnDADR` These macros will draw a negated directed adjacency downward symbol with relation spacing in math mode.

`\$u\GTSnegatedDirectedAdjacencyDownRelation/v\$`

or `\$u\GTSnegDirAdjDownRel/v\$`

or `\$u\GTSnDADR/v\$`

will yield $u \nexists v$.

See their [implementations](#) on page 30.

2.2.3 Standard incidence macros

`\GTS@i` This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`\{<mainColor>\}` If non-empty, this color will be used as main color.

`\{<secondColor>\}` If non-empty, this color will be used as second color.

`\{<drawSlash>\}` Non-empty implies that a negation will be drawn.

`\{<drawUpArrow>\}` Non-empty implies that an up-arrow will be drawn.

`\{<drawDownArrow>\}` Non-empty implies that a down-arrow will be drawn.

See its [implementation](#) on page 30.

`\GTS@j` This macro builds on top of `\GTS@i` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@i`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`.

Arguments:

- `{\mainColor}`} If non-empty, this color will be used as main color.
- `{\secondColor}`} If non-empty, this color will be used as second color.
- `{\drawSlash}`} Non-empty implies that a negation will be drawn.
- `{\drawUpArrow}`} Non-empty implies that an up-arrow will be drawn.
- `{\drawDownArrow}`} Non-empty implies that a down-arrow will be drawn.
- `{\relation}`} Non-empty implies that relation spacing will be added.

See its [implementation](#) on page 31.

`\GTSxincidence`, `\GTSxinc`, `\GTSxi` Versatile macros for incidence that can be given options.

Argument:

`[\optionsList]` Sets the following options if not empty:

`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`in` or `i`,
`out` or `o`,
`relation` or `rel` or `r`.

See their [implementations](#) on page 31.

`\GTSincidence`, `\GTSinc`, `\GTSi` These macros will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

`A.\GTSincidence/A.`
or `A.\GTSinc/A.`
or `A.\GTSi/A.`
will yield $A \downarrow A.$

`v \GTSincidence/ e`
or `v \GTSinc/ e`
or `v \GTSi/ e`
will yield $v \downarrow e.$

`$v \GTSincidence/ e$`
or `$v \GTSinc/ e$`
or `$v \GTSi/ e$`
will yield $v \downarrow e.$

See their [implementations](#) on page 32.

`\GTSincidenceRelation`, `\GTSincRel`, `\GTSiR` These macros will draw an incidence symbol with relation spacing in math mode.

`$v\GTSincidenceRelation/e$`
or `$v\GTSincRel/e$`
or `$v\GTSiR/e$`
will yield $v \downarrow e.$

See their [implementations](#) on page 32.

`\GTSnegatedIncidence`, `\GTSnegInc`, `\GTSni` These macros will draw a negated incidence symbol.

`A.\GTSnegatedIncidence/A.`

or $A.\backslash\text{GTSnegInc}/A.$

or $A.\backslash\text{GTSnI}/A.$

will yield $A.\not\vdash A.$

$v \backslash\text{GTSnegatedIncidence}/ e$

or $v \backslash\text{GTSnegInc}/ e$

or $v \backslash\text{GTSnI}/ e$

will yield $v \not\vdash e.$

$\$v \backslash\text{GTSnegatedIncidence}/ e\$$

or $\$v \backslash\text{GTSnegInc}/ e\$$

or $\$v \backslash\text{GTSnI}/ e\$$

will yield $v \not\vdash e.$

See their [implementations](#) on page 32.

$\backslash\text{GTSnegatedIncidenceRelation}$, $\backslash\text{GTSnegIncRel}$, $\backslash\text{GTSnIR}$ These macros will draw a negated incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSnegIncRel}/e\$$

or $\$v\backslash\text{GTSnIR}/e\$$

will yield $v \not\vdash e.$

See their [implementations](#) on page 32.

$\backslash\text{GTSoutIncidence}$, $\backslash\text{GTSoutInc}$, $\backslash\text{GTSoI}$ These macros will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSoutIncidence}/A.$

or $A.\backslash\text{GTSoutInc}/A.$

or $A.\backslash\text{GTSoI}/A.$

will yield $A.\uparrow A.$

$v \backslash\text{GTSoutIncidence}/ e$

or $v \backslash\text{GTSoutInc}/ e$

or $v \backslash\text{GTSoI}/ e$

will yield $v \uparrow e.$

$\$v \backslash\text{GTSoutIncidence}/ e\$$

or $\$v \backslash\text{GTSoutInc}/ e\$$

or $\$v \backslash\text{GTSoI}/ e\$$

will yield $v \uparrow e.$

See their [implementations](#) on page 33.

$\backslash\text{GTSoutIncidenceRelation}$, $\backslash\text{GTSoutIncRel}$, $\backslash\text{GTSoIR}$ These macros will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSoutIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSoutIncRel}/e\$$

or $\$v\backslash\text{GTSoIR}/e\$$

will yield $v \nrightarrow e$.

See their [implementations](#) on page 33.

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI These macros will draw a negated directed incidence upward (out(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedOutIncidence}/A.$

or $A.\backslash\text{GTSnegOutInc}/A.$

or $A.\backslash\text{GTSnOI}/A.$

will yield $A.\nrightarrow A.$

$v \backslash\text{GTSnegatedOutIncidence}/ e$

or $v \backslash\text{GTSnegOutInc}/ e$

or $v \backslash\text{GTSnOI}/ e$

will yield $v \nrightarrow e$.

$\$v \backslash\text{GTSnegatedOutIncidence}/ e\$$

or $\$v \backslash\text{GTSnegOutInc}/ e\$$

or $\$v \backslash\text{GTSnOI}/ e\$$

will yield $v \nrightarrow e$.

See their [implementations](#) on page 33.

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR These macros will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedOutIncidenceRelation}/e\$$

or $\$v\backslash\text{GTSnegOutIncRel}/e\$$

or $\$v\backslash\text{GTSnOIR}/e\$$

will yield $v \nrightarrow e$.

See their [implementations](#) on page 33.

\GTSinIncidence, \GTSinInc, \GTSiI These macros will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSinIncidence}/A.$

or $A.\backslash\text{GTSinInc}/A.$

or $A.\backslash\text{GTSiI}/A.$

will yield $A.\rightarrow A.$

$v \backslash\text{GTSinIncidence}/ e$

or $v \backslash\text{GTSinInc}/ e$

or $v \backslash\text{GTSiI}/ e$

will yield $v \rightarrow e$.

$\$v \backslash\text{GTSinIncidence}/ e\$$

or $\$v \backslash\text{GTSinInc}/ e\$$

or $\$v \backslash\text{GTSiI}/ e\$$

will yield $v \rightarrow e$.

See their [implementations](#) on page 33.

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR These macros will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`$v\GTSinIncidenceRelation/e$`

or `$v\GTSinIncRel/e$`

or `$v\GTSiIR/e$`

will yield $v \downarrow e$.

See their [implementations](#) on page 33.

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII These macros will draw a negated directed incidence downward (in(ward) incidence) symbol.

`A.\GTSnegatedInIncidence/A.`

or `A.\GTSnegInInc/A.`

or `A.\GTSnII/A.`

will yield $A.\nmid A.$

`v \GTSnegatedInIncidence/ e`

or `v \GTSnegInInc/ e`

or `v \GTSnII/ e`

will yield $v \nmid e$.

`$v \GTSnegatedInIncidence/ e$`

or `$v \GTSnegInInc/ e$`

or `$v \GTSnII/ e$`

will yield $v \nmid e$.

See their [implementations](#) on page 33.

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR These macros will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

`$v\GTSnegatedInIncidenceRelation/e$`

or `$v\GTSnegInIncRel/e$`

or `$v\GTSnIIR/e$`

will yield $v \nmid e$.

See their [implementations](#) on page 33.

2.2.4 Monochrome adjacency macros

\GTSadjacencyMonochrome, \GTSadjMono, \GTSadjMon, \GTSaM These macros will draw a monochrome adjacency symbol.

`A.\GTSadjacencyMonochrome/A.`

or `A.\GTSadjMono/A.`

or `A.\GTSadjMon/A.`

or `A.\GTSaM/A.`

will yield $A.\circ A.$

`u \GTSadjacencyMonochrome/ v`

or `u \GTSadjMono/ v`

or $\text{u} \text{ \GTSadjMon/ } \text{v}$

or $\text{u} \text{ \GTSaM/ } \text{v}$

will yield $\text{u} \circ \text{v}$.

$\text{\$u \GTSadjacencyMonochrome/ v\$}$

or $\text{\$u \GTSadjMono/ v\$}$

or $\text{\$u \GTSadjMon/ v\$}$

or $\text{\$u \GTSaM/ v\$}$

will yield $u \circ v$.

See their [implementations](#) on page 34.

\GTSadjacencyMonochromeRelation, \GTSadjMonoRel, \GTSadjMonRel, \GTSaMR These macros will draw a monochrome adjacency symbol with relation spacing in math mode.

$\text{\$u\GTSadjacencyMonochromeRelation/v\$}$

or $\text{\$u\GTSadjMonoRel/v\$}$

or $\text{\$u\GTSadjMonRel/v\$}$

or $\text{\$u\GTSaMR/v\$}$

will yield $u \circ v$.

See their [implementations](#) on page 34.

\GTSunadjacencyMonochrome, \GTSunadjMono, \GTSunadjMon, \GTSuM These macros will draw a monochrome unadjacency symbol.

$\text{A.\GTSunadjacencyMonochrome/A.}$

or $\text{A.\GTSunadjMono/A.}$

or $\text{A.\GTSunadjMon/A.}$

or $\text{A.\GTSuM/A.}$

will yield $A \circ A$.

$\text{u \GTSunadjacencyMonochrome/ v}$

or $\text{u \GTSunadjMono/ v}$

or $\text{u \GTSunadjMon/ v}$

or $\text{u \GTSuM/ v}$

will yield $u \circ v$.

$\text{\$u \GTSunadjacencyMonochrome/ v\$}$

or $\text{\$u \GTSunadjMono/ v\$}$

or $\text{\$u \GTSunadjMon/ v\$}$

or $\text{\$u \GTSuM/ v\$}$

will yield $u \circ v$.

See their [implementations](#) on page 34.

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR These macros will draw a monochrome unadjacency symbol with relation spacing in math mode.

$\text{\$u\GTSunadjacencyMonochromeRelation/v\$}$

or $\text{\$u\GTSunadjMonoRel/v\$}$

or $\$u\backslash\text{GTSunadjMonRel}/v\$$

or $\$u\backslash\text{GTSuMR}/v\$$

will yield $u \circ v$.

See their [implementations](#) on page 34.

$\backslash\text{GTSdirectedAdjacencyUpMonochrome}$, $\backslash\text{GTSdirAdjUpMono}$, $\backslash\text{GTSdirAdjUpMon}$, $\backslash\text{GTSdAUM}$ These macros will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A.\backslash\text{GTSdirectedAdjacencyUpMonochrome}/A.$

or $A.\backslash\text{GTSdirAdjUpMono}/A.$

or $A.\backslash\text{GTSdirAdjUpMon}/A.$

or $A.\backslash\text{GTSdAUM}/A.$

will yield $A.\circ A.$.

$u \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ v$

or $u \backslash\text{GTSdirAdjUpMono}/ v$

or $u \backslash\text{GTSdirAdjUpMon}/ v$

or $u \backslash\text{GTSdAUM}/ v$

will yield $u \circ v$.

$\$u \backslash\text{GTSdirectedAdjacencyUpMonochrome}/ v\$$

or $\$u \backslash\text{GTSdirAdjUpMono}/ v\$$

or $\$u \backslash\text{GTSdirAdjUpMon}/ v\$$

or $\$u \backslash\text{GTSdAUM}/ v\$$

will yield $u \circ v$.

See their [implementations](#) on page 34.

$\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}$, $\backslash\text{GTSdirAdjUpMonoRel}$, $\backslash\text{GTSdirAdjUpMonRel}$, $\backslash\text{GTSdAUMR}$ These macros will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/v\$$

or $\$u\backslash\text{GTSdirAdjUpMonoRel}/v\$$

or $\$u\backslash\text{GTSdirAdjUpMonRel}/v\$$

or $\$u\backslash\text{GTSdAUMR}/v\$$

will yield $u \circ v$.

See their [implementations](#) on page 34.

$\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}$, $\backslash\text{GTSnegDirAdjUpMono}$, $\backslash\text{GTSnegDirAdjUpMon}$, $\backslash\text{GTSnDAUM}$ These macros will draw a monochrome negated directed adjacency upward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A.$

or $A.\backslash\text{GTSnegDirAdjUpMono}/A.$

or $A.\backslash\text{GTSnegDirAdjUpMon}/A.$

or $A.\backslash\text{GTSnDAUM}/A.$

will yield $A.\circ A.$.

`u \GTSnegatedDirectedAdjacencyUpMonochrome/ v`

or `u \GTSnegDirAdjUpMono/ v`

or `u \GTSnegDirAdjUpMon/ v`

or `u \GTSnDAUM/ v`

will yield $u \overset{\circ}{\rightharpoonup} v$.

`$u \GTSnegatedDirectedAdjacencyUpMonochrome/ v$`

or `$u \GTSnegDirAdjUpMono/ v$`

or `$u \GTSnegDirAdjUpMon/ v$`

or `$u \GTSnDAUM/ v$`

will yield $u \overset{\circ}{\rightharpoonup} v$.

See their [implementations](#) on page 34.

`\GTSnegatedDirectedAdjacencyUpMonochromeRelation`, `\GTSnegDirAdjUpMonoRel`, `\GTSnegDirAdjUpMonRel`, `\GTSnDAUMR` These macros will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

`$u \GTSnegatedDirectedAdjacencyUpMonochromeRelation/ v$`

or `$u \GTSnegDirAdjUpMonoRel/ v$`

or `$u \GTSnegDirAdjUpMonRel/ v$`

or `$u \GTSnDAUMR/ v$`

will yield $u \overset{\circ}{\rightharpoonup} v$.

See their [implementations](#) on page 35.

`\GTSdirectedAdjacencyDownMonochrome`, `\GTSdirAdjDownMono`, `\GTSdirAdjDownMon`, `\GTSdADM` These macros will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

`A.\GTSdirectedAdjacencyDownMonochrome/A.`

or `A.\GTSdirAdjDownMono/A.`

or `A.\GTSdirAdjDownMon/A.`

or `A.\GTSdADM/A.`

will yield $A \overset{\circ}{\searrow} A$.

`u \GTSdirectedAdjacencyDownMonochrome/ v`

or `u \GTSdirAdjDownMono/ v`

or `u \GTSdirAdjDownMon/ v`

or `u \GTSdADM/ v`

will yield $u \overset{\circ}{\searrow} v$.

`$u \GTSdirectedAdjacencyDownMonochrome/ v$`

or `$u \GTSdirAdjDownMono/ v$`

or `$u \GTSdirAdjDownMon/ v$`

or `$u \GTSdADM/ v$`

will yield $u \overset{\circ}{\searrow} v$.

See their [implementations](#) on page 35.

\GTSdirectedAdjacencyDownMonochromeRelation, \GTSdirAdjDownMonoRel, \GTSdirAdjDownMonRel, \GTSdADMR These macros will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}/v\$$
or $\$u\backslash\text{GTSdirAdjDownMonoRel}/v\$$
or $\$u\backslash\text{GTSdirAdjDownMonRel}/v\$$
or $\$u\backslash\text{GTSdADMR}/v\$$

will yield $u \Downarrow v$.

See their [implementations](#) on page 35.

\GTSnegatedDirectedAdjacencyDownMonochrome, \GTSnegDirAdjDownMono, \GTSnegDirAdjDownMon, \GTSnDADM These macros will draw a monochrome negated directed adjacency downward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/A.$
or $A.\backslash\text{GTSnegDirAdjDownMono}/A.$
or $A.\backslash\text{GTSnegDirAdjDownMon}/A.$
or $A.\backslash\text{GTSnDADM}/A.$

will yield $A.\Downarrow A.$

$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v$
or $u \backslash\text{GTSnegDirAdjDownMono}/ v$
or $u \backslash\text{GTSnegDirAdjDownMon}/ v$
or $u \backslash\text{GTSnDADM}/ v$

will yield $u \Downarrow v$.

$\$u \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/ v\$$
or $\$u \backslash\text{GTSnegDirAdjDownMono}/ v\$$
or $\$u \backslash\text{GTSnegDirAdjDownMon}/ v\$$
or $\$u \backslash\text{GTSnDADM}/ v\$$

will yield $u \Downarrow v$.

See their [implementations](#) on page 35.

\GTSnegatedDirectedAdjacencyDownMonochromeRelation, \GTSnegDirAdjDownMonoRel, \GTSnegDirAdjDownMonRel, \GTSnDADMR These macros will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

$\$u\backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}/v\$$
or $\$u\backslash\text{GTSnegDirAdjDownMonoRel}/v\$$
or $\$u\backslash\text{GTSnegDirAdjDownMonRel}/v\$$
or $\$u\backslash\text{GTSnDADMR}/v\$$

will yield $u \Downarrow v$.

See their [implementations](#) on page 35.

2.2.5 Monochrome incidence macros

\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM These macros will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSincidenceMonochrome}/A.$

or $A.\backslash\text{GTSincMono}/A.$

or $A.\backslash\text{GTSincMon}/A.$

or $A.\backslash\text{GTSiM}/A.$

will yield $A.\downarrow A.$

$v \backslash\text{GTSincidenceMonochrome}/ e$

or $v \backslash\text{GTSincMono}/ e$

or $v \backslash\text{GTSincMon}/ e$

or $v \backslash\text{GTSiM}/ e$

will yield $v \downarrow e$

$\$v \backslash\text{GTSincidenceMonochrome}/ e\$$

or $\$v \backslash\text{GTSincMono}/ e\$$

or $\$v \backslash\text{GTSincMon}/ e\$$

or $\$v \backslash\text{GTSiM}/ e\$$

will yield $v \downarrow e$

See their [implementations](#) on page 36.

$\backslash\text{GTSincidenceMonochromeRelation}$, $\backslash\text{GTSincMonoRel}$, $\backslash\text{GTSincMonRel}$, $\backslash\text{GTSiMR}$ These macros will draw a monochrome incidence symbol with relation spacing in math mode.

$\$v\backslash\text{GTSincidenceMonochromeRelation}/e\$$

or $\$v\backslash\text{GTSincMonoRel}/e\$$

or $\$v\backslash\text{GTSincMonRel}/e\$$

or $\$v\backslash\text{GTSiMR}/e\$$

will yield $v \downarrow e$

See their [implementations](#) on page 36.

$\backslash\text{GTSnegatedIncidenceMonochrome}$, $\backslash\text{GTSnegIncMono}$, $\backslash\text{GTSnegIncMon}$, $\backslash\text{GTSnIM}$ These macros will draw a monochrome negated incidence symbol.

$A.\backslash\text{GTSnegatedIncidenceMonochrome}/A.$

or $A.\backslash\text{GTSnegIncMono}/A.$

or $A.\backslash\text{GTSnegIncMon}/A.$

or $A.\backslash\text{GTSnIM}/A.$

will yield $A.\not\downarrow A.$

$v \backslash\text{GTSnegatedIncidenceMonochrome}/ e$

or $v \backslash\text{GTSnegIncMono}/ e$

or $v \backslash\text{GTSnegIncMon}/ e$

or $v \backslash\text{GTSnIM}/ e$

will yield $v \not\downarrow e$

$\$v \backslash\text{GTSnegatedIncidenceMonochrome}/ e\$$

or $\$v \backslash\text{GTSnegIncMono}/ e\$$

or $\$v \backslash\text{GTSnegIncMon}/ e\$$

or $\$v \backslash\text{GTSnIM}/ e\$$

will yield $v \not\downarrow e$

See their [implementations](#) on page 36.

\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR These macros will draw a monochrome negated incidence symbol with relation spacing in math mode.

`\$v\GTSnegatedIncidenceMonochromeRelation/e\$`

or `\$v\GTSnegIncMonoRel/e\$`

or `\$v\GTSnegIncMonRel/e\$`

or `\$v\GTSnIMR/e\$`

will yield $v \nparallel e$.

See their [implementations](#) on page 36.

\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSoIM These macros will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`A.\GTSoutIncidenceMonochrome/A.`

or `A.\GTSoutIncMono/A.`

or `A.\GTSoutIncMon/A.`

or `A.\GTSoIM/A.`

will yield $A \uparrow A$.

`v \GTSoutIncidenceMonochrome/ e`

or `v \GTSoutIncMono/ e`

or `v \GTSoutIncMon/ e`

or `v \GTSoIM/ e`

will yield $v \uparrow e$.

`\$v \GTSoutIncidenceMonochrome/ e\$`

or `\$v \GTSoutIncMono/ e\$`

or `\$v \GTSoutIncMon/ e\$`

or `\$v \GTSoIM/ e\$`

will yield $v \uparrow e$.

See their [implementations](#) on page 36.

\GTSoutIncidenceMonochromeRelation, \GTSoutIncMonoRel, \GTSoutIncMonRel, \GTSoIMR These macros will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`\$v\GTSoutIncidenceMonochromeRelation/e\$`

or `\$v\GTSoutIncMonoRel/e\$`

or `\$v\GTSoutIncMonRel/e\$`

or `\$v\GTSoIMR/e\$`

will yield $v \uparrow e$.

See their [implementations](#) on page 36.

\GTSnegatedOutIncidenceMonochrome, \GTSnegOutIncMono, \GTSnegOutIncMon, \GTSnOIM These macros will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

`A.\GTSnegatedOutIncidenceMonochrome/A.`

or `A.\GTSnegOutIncMono/A.`

or `A.\GTSnegOutIncMon/A.`

or `A.\GTSnOIM/A.`

will yield $A \nrightarrow A$.

`v \GTSnegatedOutIncidenceMonochrome/ e`

or `v \GTSnegOutIncMono/ e`

or `v \GTSnegOutIncMon/ e`

or `v \GTSnOIM/ e`

will yield $v \nrightarrow e$.

`$v \GTSnegatedOutIncidenceMonochrome/ e$`

or `$v \GTSnegOutIncMono/ e$`

or `$v \GTSnegOutIncMon/ e$`

or `$v \GTSnOIM/ e$`

will yield $v \nrightarrow e$.

See their [implementations](#) on page 36.

`\GTSnegatedOutIncidenceMonochromeRelation`, `\GTSnegOutIncMonoRel`, `\GTSnegOutIncMonRel`, `\GTSnOIMR` These macros will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

`$v\GTSnegatedOutIncidenceMonochromeRelation/e$`

or `$v\GTSnegOutIncMonoRel/e$`

or `$v\GTSnegOutIncMonRel/e$`

or `$v\GTSnOIMR/e$`

will yield $v \nrightarrow e$.

See their [implementations](#) on page 37.

`\GTSinIncidenceMonochrome`, `\GTSinIncMono`, `\GTSinIncMon`, `\GTSiIM` These macros will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`A.\GTSinIncidenceMonochrome/A.`

or `A.\GTSinIncMono/A.`

or `A.\GTSinIncMon/A.`

or `A.\GTSiIM/A.`

will yield $A \rightarrow A$.

`v \GTSinIncidenceMonochrome/ e`

or `v \GTSinIncMono/ e`

or `v \GTSinIncMon/ e`

or `v \GTSiIM/ e`

will yield $v \rightarrow e$.

`$v \GTSinIncidenceMonochrome/ e$`

or $\$v \backslash \text{GTSinIncMono}/ e\$$
or $\$v \backslash \text{GTSinIncMon}/ e\$$
or $\$v \backslash \text{GTSiIM}/ e\$$

will yield $v \Downarrow e$.

See their [implementations](#) on page 37.

$\backslash \text{GTSinIncidenceMonochromeRelation}$, $\backslash \text{GTSinIncMonoRel}$, $\backslash \text{GTSinIncMonRel}$, $\backslash \text{GTSiIMR}$
These macros will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v \backslash \text{GTSinIncidenceMonochromeRelation}/ e\$$
or $\$v \backslash \text{GTSinIncMonoRel}/ e\$$
or $\$v \backslash \text{GTSinIncMonRel}/ e\$$
or $\$v \backslash \text{GTSiIMR}/ e\$$

will yield $v \Downarrow e$.

See their [implementations](#) on page 37.

$\backslash \text{GTSnegatedInIncidenceMonochrome}$, $\backslash \text{GTSnegInIncMono}$, $\backslash \text{GTSnegInIncMon}$, $\backslash \text{GTSnIIM}$
These macros will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

$A. \backslash \text{GTSnegatedInIncidenceMonochrome}/ A.$
or $A. \backslash \text{GTSnegInIncMono}/ A.$
or $A. \backslash \text{GTSnegInIncMon}/ A.$
or $A. \backslash \text{GTSnIIM}/ A.$

will yield $A. \Downarrow A.$.

$v \backslash \text{GTSnegatedInIncidenceMonochrome}/ e$
or $v \backslash \text{GTSnegInIncMono}/ e$
or $v \backslash \text{GTSnegInIncMon}/ e$
or $v \backslash \text{GTSnIIM}/ e$

will yield $v \Downarrow e$.

$\$v \backslash \text{GTSnegatedInIncidenceMonochrome}/ e\$$
or $\$v \backslash \text{GTSnegInIncMono}/ e\$$
or $\$v \backslash \text{GTSnegInIncMon}/ e\$$
or $\$v \backslash \text{GTSnIIM}/ e\$$

will yield $v \Downarrow e$.

See their [implementations](#) on page 37.

$\backslash \text{GTSnegatedInIncidenceMonochromeRelation}$, $\backslash \text{GTSnegInIncMonoRel}$, $\backslash \text{GTSnegInIncMonRel}$, $\backslash \text{GTSnIIMR}$
These macros will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v \backslash \text{GTSnegatedInIncidenceMonochromeRelation}/ e\$$
or $\$v \backslash \text{GTSnegInIncMonoRel}/ e\$$
or $\$v \backslash \text{GTSnegInIncMonRel}/ e\$$
or $\$v \backslash \text{GTSnIIMR}/ e\$$

will yield $v \not\bowtie e$.

See their [implementations](#) on page 37.

2.2.6 Meta-logic macros

\GTSmetaNot, \GTSmN These macros will draw a meta-not or relation-not symbol.

A.\GTSmetaNot/A.

or A.\GTSmN/A.

will yield $A.\mathrel{\mathbin{\boxtimes}}A$.

u \GTSmetaNot/\GTSdAU/ v

or u \GTSmN/\GTSdAU/ v

will yield $u \mathrel{\mathbin{\boxtimes}} v$.

\$u \GTSmetaNot/ \GTSdAU/ v\$

or \$u \GTSmN/ \GTSdAU/ v\$

will yield $u \mathrel{\mathbin{\boxtimes}} v$.

See their [implementations](#) on page 37.

\GTSmetaAnd, \GTSmA These macros will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

A.\GTSmetaAnd/A.

or A.\GTSmA/A.

will yield $A.\mathrel{\mathbin{\boxtimes}}A$.

u \GTSdAD/\GTSmetaAnd/\GTSdAU/ v

or u \GTSdAD/\GTSmA/\GTSdAU/ v

will yield $u \mathrel{\mathbin{\boxtimes}} v$.

\$u \GTSdAD/ \GTSmetaAnd/ \GTSdAU/ v\$

or \$u \GTSdAD/ \GTSmA/ \GTSdAU/ v\$

will yield $u \mathrel{\mathbin{\boxtimes}} v$.

See their [implementations](#) on page 38.

\GTSmetaAndRelation, \GTSmetaAndRel, \GTSmAR These macros will draw a meta-and symbol with relation spacing in math mode.

\$u \GTSdAD/\GTSmetaAndRelation/\GTSdAU/ v\$

or \$u \GTSdAD/\GTSmetaAndRel/\GTSdAU/ v\$

or \$u \GTSdAD/\GTSmAR/\GTSdAU/ v\$

will yield $u \mathrel{\mathbin{\boxtimes}} v$.

See their [implementations](#) on page 38.

\GTSmetaOr, \GTSmO These macros will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

A.\GTSmetaOr/A.

or A.\GTSmO/A.

will yield $A.\mathrel{\mathbin{\boxtimes}}A$.

u \GTSdAD/\GTSmetaOr/\GTSdAU/ v

or u \GTSdAD/\GTSmO/\GTSdAU/ v

will yield $u \mathrel{\mathbin{\boxtimes}} v$.

\$u \GTSdAD/ \GTSmetaOr/ \GTSdAU/ v\$

or \$u \GTSdAD/ \GTSmO/ \GTSdAU/ v\$

will yield $u \overset{\circ}{\underset{\circ}{\vee}} v$.

See their [implementations](#) on page 38.

\GTSMetaOrRelation, \GTSMetaOrRel, \GTSMOR These macros will draw a meta-or symbol with relation spacing in math mode.

\$u \GTSDAD/\GTSMetaOrRelation/\GTSdAU/ v\$

or \$u \GTSDAD/\GTSMetaOrRel/\GTSdAU/ v\$

or \$u \GTSDAD/\GTSMOR/\GTSdAU/ v\$

will yield $u \overset{\circ}{\underset{\circ}{\vee}} v$.

See their [implementations](#) on page 38.

\GTSxmetallogic, \GTSxmi Versatile macros for meta-logic symbols that can be given options.

Argument:

[*<optionsList>*] Sets the following options if not empty:

not or n,

and or a,

or or o,

relation or rel or r.

Of course first three options are incompatible, and the fourth is compatible only with the second and third options.

See their [implementations](#) on page 38.

\GTSx One macro to rule them all. Versatile macro for all symbols that can be given options, and a symbol type mandatory switch.

Argument:

[*<optionsList>*] Sets the following options if not empty:

- If $\{\langle symbolType \rangle\}$ below is a, see \GTSxa:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.
- If $\{\langle symbolType \rangle\}$ below is i, see \GTSxi:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.
- If $\{\langle symbolType \rangle\}$ below is m, see \GTSxmi:
not or n,
and or a,
or or o,
relation or rel or r.

$\{\langle symbolType \rangle\}$ a mandatory switch; it can take value a for adjacency, i for incidence, m for meta-logic.

See its [implementation](#) on page 39.

2.3 Good practices

As is customary in algebra and logic, the conjunction/multiplication can be replaced by concatenation, and has higher priority than disjunction. So the following formulas are equivalent; we recommend using the shortest one:

Tournament adjacency:

$$\forall u, v, u \text{ --- } \overset{\text{m}}{\nabla} \text{ --- } v$$

is preferred over

$$\forall u, v, u \text{ } \begin{array}{c} \textcircled{\text{m}} \\ \textcircled{\wedge} \end{array} \textcircled{\text{m}} \textcircled{\vee} \textcircled{\text{m}} \textcircled{\wedge} \textcircled{\text{m}} v,$$
$$\text{or } \forall u, v, u \left(\bigotimes_{\bigotimes}^{\bigotimes} \bigwedge \bigotimes \right)^{\bigotimes} \bigvee \left(\bigotimes_{\bigotimes}^{\bigotimes} \bigwedge \bigotimes \right)^{\bigotimes} v,$$

or even $\forall u, v, ((u \overset{\circ}{\times} v) \wedge (u \overset{\circ}{\times} v)) \vee ((u \overset{\circ}{\times} v) \wedge (u \overset{\circ}{\times} v))$.

Note the use of macros without Relation suffix above, and adding a single `\mathrel{}` around the composition of all relations:

$$\backslash\mathrm{mathrel}\{\backslash\mathrm{GTSdAD}/\backslash\mathrm{GTSnDAU}/\backslash\mathrm{GTSmO}/\backslash\mathrm{GTSnDAD}/\backslash\mathrm{GTSdAU}/\}.$$

This is the spacing practice that we recommend, although for very long compositions, it may be quite unreadable, and then spacing everything with `Relation` could/should be preferred.

We also recommend using brackets when the composition reflects all of the adjacencies between two vertices:

Tournament adjacency:

$$\forall u, v, u \left[\begin{array}{c} \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \\ \textcircled{\circ} \textcircled{\circ} \textcircled{\circ} \end{array} \textcircled{\circ} \textcircled{\circ} \right] v.$$

Tournament adjacency with maybe some colored edge also, who knows?:

$$\forall u, v, u \text{ } \begin{array}{c} \circ \\ \vdots \\ \circ \end{array} \text{ } \begin{array}{c} \circ \\ \vdots \\ \circ \end{array} \text{ } \begin{array}{c} \circ \\ \vdots \\ \circ \end{array} \text{ } \begin{array}{c} \circ \\ \vdots \\ \circ \end{array} \text{ } v.$$

Directed graph:

[illegible]

Oriented graph:

[illegible]

Finally, we recommend prefix notation without “big” symbol for a sequence of consecutive ors or ands (with parentheses if needed):

$$\exists u, v, u \left[\overset{\text{m}}{\underset{\text{V}}{\vee}} \begin{array}{ccc} \text{red} & \text{green} & \text{blue} \\ \text{red} & \text{green} & \text{blue} \end{array} \right] v.$$
$$\exists v, e, v \left[\overset{\text{m}}{\underset{\text{V}}{\text{V}}} \text{ } \text{ } \text{ } \right] e.$$

3 Implementation

Below, we try to avoid repeating each macro description, and we give a hyperlink plus a page number reference of its description, after macro name(s). But if the description is only one sentence, we do repeat it here, because it makes no sense to have to go back and forth in the text for only one sentence.

```

1 (*package)
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{keyval}
6 \RequirePackage{fdsymbol}
7 \RequirePackage{lua-unicode-math}
8 \RequirePackage{xcolor}
9
10 \def\GTS@mainColor{black}
11 \def\GTS@secondColor{gray}
12 \def\GTS@mainColorBackup{black}
13 \def\GTS@secondColorBackup{gray}
14 \def\GTS@drawSlash{}
15 \def\GTS@drawUpArrow{}
16 \def\GTS@drawDownArrow{}
17
18 \GTS@newcommand See its description on page 7.
19 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
20   \@ifdefinable{#1}{\def#1/{#2}}
21 }%

```

`\GTS@newprotectedcommand` See its [description](#) on page 7.

```
21 \ifdefinable{\GTS@newprotectedcommand}{\def\GTS@newprotectedcommand#1#2{%
22   \ifdefinable{#1}{\protected\def#1/{#2}}
23 }}
```

`\GTSsetMainColor` See its [description](#) on page 7.

```
24 \newcommand{\GTSsetMainColor}[1]{%
25   \edef\GTS@mainColorTemp{#1}%
26   \edef\GTS@mainColorBackup{\GTS@mainColor}%
27   \edef\GTS@mainColor{\GTS@mainColorTemp}%
28 }
```

`\GTSsetSecondColor` See its [description](#) on page 7.

```
29 \newcommand{\GTSsetSecondColor}[1]{%
30   \edef\GTS@secondColorTemp{#1}%
31   \edef\GTS@secondColorBackup{\GTS@secondColor}%
32   \edef\GTS@secondColor{\GTS@secondColorTemp}%
33 }
```

`\GTS@a` See its [description](#) on page 7.

```
34 \newcommand{\GTS@a}[5]{%
35   \if#1\empty\else%
36     \GTSsetMainColor{#1}%
37     \fi%
38     \if#2\empty\else%
39       \GTSsetSecondColor{#2}%
40       \fi%
41     \text{\ensuremath{%
42       \raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
43         %
44         \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
45           $\mathcolor{\GTS@secondColor}{\circ}$%
46         }}%
47         \hspace{-0.39ex}%
48         \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
49           $\mathcolor{\GTS@secondColor}{\bullet}$%
50         }}%
51         \hspace{-0.12ex}%
52         \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
53           $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
54         }}%
55         \hspace{-0.40ex}%
56         \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
57           $\mathcolor{\GTS@secondColor}{\circ}$%
58         }}%
59         \hspace{-0.39ex}%
60         \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
61           $\mathcolor{\GTS@secondColor}{\bullet}$%
62         }}%
63         \hspace{0.27ex}%

```

if `\GTS@drawUpArrow` empty else

```
64 \if#4\empty\else%
65   \hspace{-0.65ex}%
66   \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
67     \textcolor{\GTS@mainColor}{\textasciicircum}%
68   }}%
69 \fi%
```

if `\GTS@drawDownArrow` empty else

```
70 \if#5\empty\else%
71   \hspace{-0.65ex}%

```

```

72 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
73 \textcolor{\GTS@mainColor}{\textasciicircum}%
74 }}%
75 \fi%

```

if GTS@drawSlash empty else

```

76 \if#3\empty\else%
77 \hspace{-0.65ex}%
78 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
79 \textcolor{black}{/}%
80 }}%
81 \fi%
82 %
83 }}%
84 }}%
85 \if#1\empty\else%
86 \GTSsetMainColor{\GTS@mainColorBackup}%
87 \fi%
88 \if#2\empty\else%
89 \GTSsetSecondColor{\GTS@secondColorBackup}%
90 \fi%
91 }

```

\GTS@b See its [description](#) on page 7.

```

92 \newcommand{\GTS@b}[6]{%
93 \if#61%
94 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
95 \else%
96 \GTS@a{#1}{#2}{#3}{#4}{#5}%
97 \fi%
98 }

```

\GTSxadjacency, **\GTSxadj**, **\GTSxa** See their [description](#) on page 8.

```

99 \define@key{\GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
100 \define@key{\GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
101 \define@key{\GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
102 \define@key{\GTSxa}{mC}{\def\GTS@@mainColor{#1}}
103 \define@key{\GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
104 \define@key{\GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
105 \define@key{\GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
106 \define@key{\GTSxa}{sC}{\def\GTS@@secondColor{#1}}
107 \define@key{\GTSxa}{monochrome}[1]{%
108 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
109 \define@key{\GTSxa}{mono}[1]{%
110 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
111 \define@key{\GTSxa}{mon}[1]{%
112 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
113 \define@key{\GTSxa}{m}[1]{%
114 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
115 \define@key{\GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
116 \define@key{\GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
117 \define@key{\GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
118 \define@key{\GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
119 \define@key{\GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
120 \define@key{\GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
121 \define@key{\GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
122 \define@key{\GTSxa}{relation}[1]{\def\GTS@@relation{1}}
123 \define@key{\GTSxa}{rel}[1]{\def\GTS@@relation{1}}
124 \define@key{\GTSxa}{r}[1]{\def\GTS@@relation{1}}
125
126 \newcommand{\GTSxa}[1][ ]{%
127 \def\GTS@@mainColor{}%

```

```

128 \def\GTS@@secondColor{%
129 \def\GTS@@drawSlash{%
130 \def\GTS@@drawUpArrow{%
131 \def\GTS@@drawDownArrow{%
132 \def\GTS@@relation{%
133 \setkeys{GTSxa}{#1}%
134 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
135 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
136 }
137
138 \newcommand{\GTSxadj}[1][\GTSxa{#1}]
139
140 \newcommand{\GTSxadjacency}[1][\GTSxa{#1}]

```

\GTSadjacency, \GTSadj, \GTSa These macros will draw an adjacency symbol. See their [description](#) and usage examples on page 8.

```

141 \GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}
142 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}
143 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}

```

\GTSadjacencyRelation, \GTSadjRel, \GTSaR These macros will draw an adjacency symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 8.

```

144 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
145 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
146 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

\GTSunadjacency, \GTSunadj, \GTSu These macros will draw an unadjacency symbol. See their [description](#) and usage examples on page 8.

```

147 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}}
148 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}}
149 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}}

```

\GTSunadjacencyRelation, \GTSunadjRel, \GTSuR These macros will draw an unadjacency symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 9.

```

150 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
151 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
152 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

\GTSdirectedAdjacencyUp, \GTSdirAdjUp, \GTSdAU These macros will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

See their [description](#) and usage examples on page 9.

```

153 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}}
154 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}}
155 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}}

```

\GTSdirectedAdjacencyUpRelation, \GTSdirAdjUpRel, \GTSdAUR These macros will draw a directed adjacency upward symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 9.

```

156 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
157 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
158 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

\GTSnegatedDirectedAdjacencyUp, \GTSnegDirAdjUp, \GTSnDAU These macros will draw a negated directed adjacency upward symbol.

See their [description](#) and usage examples on page 10.

```

159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}}
160 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}}
161 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}}

```

\GTSnegatedDirectedAdjacencyUpRelation, \GTSnegDirAdjUpRel, \GTSnDAUR These macros will draw a negated directed adjacency upward symbol with relation spacing in math mode. See their [description](#) and usage examples on page 10.

```
162 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
163 \mathrel{\GTSnDAU/}%
164 }
165 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
166 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}
```

\GTSdirectedAdjacencyDown, \GTSdirAdjDown, \GTSdAD These macros will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc. See their [description](#) and usage examples on page 10.

```
167 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}}
168 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}}
169 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}}
```

\GTSdirectedAdjacencyDownRelation, \GTSdirAdjDownRel, \GTSdADR These macros will draw a directed adjacency downward symbol with relation spacing in math mode. See their [description](#) and usage examples on page 10.

```
170 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
171 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
172 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}
```

\GTSnegatedDirectedAdjacencyDown, \GTSnegDirAdjDown, \GTSnDAD These macros will draw a negated directed adjacency downward symbol. See their [description](#) and usage examples on page 11.

```
173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{1}}
174 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{1}}
175 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{1}}
```

\GTSnegatedDirectedAdjacencyDownRelation, \GTSnegDirAdjDownRel, \GTSnDADR These macros will draw a negated directed adjacency downward symbol with relation spacing in math mode. See their [description](#) and usage examples on page 11.

```
176 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
177 \mathrel{\GTSnDAD/}%
178 }
179 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
180 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}
```

\GTS@i See its [description](#) on page 11.

```
181 \newcommand{\GTS@i}[5]{%
182 \if#1\empty\else%
183 \GTSsetMainColor{#1}%
184 \fi%
185 \if#2\empty\else%
186 \GTSsetSecondColor{#2}%
187 \fi%
188 \text{\ensuremath{%
189 \raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
190 %
191 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
192 $\mathcolor{\GTS@secondColor}{\circ}$%
193 }}%
194 \hspace{-0.39ex}%
195 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
196 $\mathcolor{\GTS@secondColor}{\bullet}$%
197 }}%
198 \hspace{-0.12ex}%
199 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
200 %
201 \hspace{-0.39ex}%
202 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
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```

```

200 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
201 }%
202 \hspace{-0.18ex}%
203 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
204 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
205 }%
206 \hspace{0.2ex}%
if GTS@drawUpArrow empty else
207 \if#4\empty\else%
208 \hspace{-0.65ex}%
209 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
210 \textcolor{\GTS@secondColor}{\textasciicircum}%
211 }%
212 \fi%
if GTS@drawDownArrow empty else
213 \if#5\empty\else%
214 \hspace{-0.65ex}%
215 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
216 \textcolor{\GTS@secondColor}{\textasciicircum}%
217 }%
218 \fi%
if GTS@drawSlash empty else
219 \if#3\empty\else%
220 \hspace{-0.65ex}%
221 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
222 \textcolor{black}{/}%
223 }%
224 \fi%
225 %
226 }%
227 }%
228 \if#1\empty\else%
229 \GTSsetMainColor{\GTS@mainColorBackup}%
230 \fi%
231 \if#2\empty\else%
232 \GTSsetSecondColor{\GTS@secondColorBackup}%
233 \fi%
234 }

```

\GTS@j See its [description](#) on page 11.

```

235 \newcommand{\GTS@j}[6]{%
236 \if#61%
237 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
238 \else%
239 \GTS@i{#1}{#2}{#3}{#4}{#5}%
240 \fi%
241 }

```

\GTSxincidence, \GTSxinc, \GTSxi See their [description](#) on page 12.

```

242 \define@key{\GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
243 \define@key{\GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
244 \define@key{\GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
245 \define@key{\GTSxi}{mC}{\def\GTS@@mainColor{#1}}
246 \define@key{\GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
247 \define@key{\GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
248 \define@key{\GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
249 \define@key{\GTSxi}{sC}{\def\GTS@@secondColor{#1}}
250 \define@key{\GTSxi}{monochrome}[1]%
251 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
252 \define@key{\GTSxi}{mono}[1]%

```

```

253 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
254 \define@key{GTSxi}{mon}[1]%
255 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
256 \define@key{GTSxi}{m}[1]%
257 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
258 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
259 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
260 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
261 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
262 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
263 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
264 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
265 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
266 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
267 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
268
269 \newcommand{\GTSxi}[1][{}]{%
270 \def\GTS@@mainColor{}}%
271 \def\GTS@@secondColor{}}%
272 \def\GTS@@drawSlash{}}%
273 \def\GTS@@drawUpArrow{}}%
274 \def\GTS@@drawDownArrow{}}%
275 \def\GTS@@relation{}}%
276 \setkeys{GTSxi}{#1}%
277 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
278 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
279 }
280
281 \newcommand{\GTSxinc}[1][{}]{\GTSxi[#1]}
282
283 \newcommand{\GTSxincidence}[1][{}]{\GTSxi[#1]}

```

\GTSincidence, \GTSinc, \GTSi These macros will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

See their [description](#) and usage examples on page 12.

```

284 \GTS@newcommand{\GTSincidence}{\GTSi{}{}{}{}}
285 \GTS@newcommand{\GTSinc}{\GTSi{}{}{}{}}
286 \GTS@newcommand{\GTSi}{\GTSi{}{}{}{}}

```

\GTSincidenceRelation, \GTSincRel, \GTSiR These macros will draw an incidence symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 12.

```

287 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
288 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
289 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

```

\GTSnegatedIncidence, \GTSnegInc, \GTSnI These macros will draw a negated incidence symbol. See their [description](#) and usage examples on page 12.

```

290 \GTS@newcommand{\GTSnegatedIncidence}{\GTSi{}{}{1}{}}
291 \GTS@newcommand{\GTSnegInc}{\GTSi{}{}{1}{}}
292 \GTS@newcommand{\GTSnI}{\GTSi{}{}{1}{}}

```

\GTSnegatedIncidenceRelation, \GTSnegIncRel, \GTSnIR These macros will draw a negated incidence symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 13.

```

293 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
294 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
295 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

```

\GTSoutIncidence, \GTSoutInc, \GTSOI These macros will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

See their [description](#) and usage examples on page 13.

```
296 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{}{1}{}}
297 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{}{1}{}}
298 \GTS@newcommand{\GTSOI}{\GTS@i{}{}{}{1}{}}
```

\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSOIR These macros will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 13.

```
299 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSOI/}}
300 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSOI/}}
301 \GTS@newcommand{\GTSOIR}{\mathrel{\GTSOI/}}
```

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI These macros will draw a negated directed incidence upward (out(ward) incidence) symbol.

See their [description](#) and usage examples on page 14.

```
302 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}}
303 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}}
304 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}}
```

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR These macros will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 14.

```
305 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
306 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
307 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}
```

\GTSinIncidence, \GTSinInc, \GTSiI These macros will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

See their [description](#) and usage examples on page 14.

```
308 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{1}{}}
309 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{1}{}}
310 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{1}{}}
```

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR These macros will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 15.

```
311 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
312 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
313 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}
```

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII These macros will draw a negated directed incidence downward (in(ward) incidence) symbol.

See their [description](#) and usage examples on page 15.

```
314 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{1}{}}
315 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{1}{}}
316 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{1}{}}
```

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR These macros will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 15.

```
317 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
318 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
319 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}
```

\GTSadjacencyMonochrome, \GTSadjMono, \GTSadjMon, \GTSaM These macros will draw a monochrome adjacency symbol.

See their [description](#) and usage examples on page 15.

```
320 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}}{}{}
321 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}}{}{}
322 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}}{}{}
323 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}}{}{}

```

\GTSadjacencyMonochromeRelation, \GTSadjMonoRel, \GTSadjMonRel, \GTSaMR These macros will draw a monochrome adjacency symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 16.

```
324 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
325 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
326 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
327 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}

```

\GTSunadjacencyMonochrome, \GTSunadjMono, \GTSunadjMon, \GTSuM These macros will draw a monochrome unadjacency symbol.

See their [description](#) and usage examples on page 16.

```
328 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}}{}{}
329 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}}{}{}
330 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}}{}{}
331 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}}{}{}

```

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR

These macros will draw a monochrome unadjacency symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 16.

```
332 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
333 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
334 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
335 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}

```

\GTSdirectedAdjacencyUpMonochrome, \GTSdirAdjUpMono, \GTSdirAdjUpMon, \GTSdAUM These macros will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

See their [description](#) and usage examples on page 17.

```
336 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
337 \GTS@a{black}{black}{1}}{}%
338 }
339 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{1}}{}{}
340 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{1}}{}{}
341 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{1}}{}{}

```

\GTSdirectedAdjacencyUpMonochromeRelation, \GTSdirAdjUpMonoRel, \GTSdirAdjUpMonRel, \GTSdAUMR These macros will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 17.

```
342 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
343 \mathrel{\GTSdAUM/}}{}
344 }
345 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
346 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
347 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}

```

\GTSnegatedDirectedAdjacencyUpMonochrome, \GTSnegDirAdjUpMono, \GTSnegDirAdjUpMon, \GTSnDAUM These macros will draw a monochrome negated directed adjacency upward symbol.

See their [description](#) and usage examples on page 17.

```
348 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%

```

```

349 \GTS@a{black}{black}{1}{1}{}%
350 }
351 \GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}}
352 \GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}}
353 \GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}}

```

\GTSnegatedDirectedAdjacencyUpMonochromeRelation, \GTSnegDirAdjUpMonoRel, \GTSnegDirAdjUpMonRel, \GTSnDAUMR These macros will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 18.

```

354 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
355 \mathrel{\GTSnDAUM/}%
356 }
357 \GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}
358 \GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}
359 \GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}

```

\GTSdirectedAdjacencyDownMonochrome, \GTSdirAdjDownMono, \GTSdirAdjDownMon, \GTSdADM These macros will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

See their [description](#) and usage examples on page 18.

```

360 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
361 \GTS@a{black}{black}{-}{-}{1}%
362 }
363 \GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{-}{-}{1}}
364 \GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{-}{-}{1}}
365 \GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{-}{-}{1}}

```

\GTSdirectedAdjacencyDownMonochromeRelation, \GTSdirAdjDownMonoRel, \GTSdirAdjDownMonRel, \GTSdADMR These macros will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 19.

```

366 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
367 \mathrel{\GTSdADM/}%
368 }
369 \GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}
370 \GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}
371 \GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}

```

\GTSnegatedDirectedAdjacencyDownMonochrome, \GTSnegDirAdjDownMono, \GTSnegDirAdjDownMon, \GTSnDADM These macros will draw a monochrome negated directed adjacency downward symbol.

See their [description](#) and usage examples on page 19.

```

372 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
373 \GTS@a{black}{black}{1}{-}{1}%
374 }
375 \GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{-}{1}}
376 \GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{-}{1}}
377 \GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{-}{1}}

```

\GTSnegatedDirectedAdjacencyDownMonochromeRelation, \GTSnegDirAdjDownMonoRel, \GTSnegDirAdjDownMonRel, \GTSnDADMR These macros will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 19.

```

378 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
379 \mathrel{\GTSnDADM/}%
380 }
381 \GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}
382 \GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}
383 \GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}

```

\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM These macros will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

See their [description](#) and usage examples on page 19.

```
384 \GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{-}{-}{-}}
385 \GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{-}{-}{-}}
386 \GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{-}{-}{-}}
387 \GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{-}{-}{-}}
```

\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR These macros will draw a monochrome incidence symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 20.

```
388 \GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}
389 \GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}
390 \GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}
391 \GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}
```

\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM These macros will draw a monochrome negated incidence symbol.

See their [description](#) and usage examples on page 20.

```
392 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
393 \GTS@i{black}{black}{1}{-}{-}%
394 }
395 \GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{-}{-}}
396 \GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{-}{-}}
397 \GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{-}{-}}
```

\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR These macros will draw a monochrome negated incidence symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 21.

```
398 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}
399 \GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}
400 \GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}
401 \GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}
```

\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSoIM These macros will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

See their [description](#) and usage examples on page 21.

```
402 \GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{-}{1}{-}}
403 \GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{-}{1}{-}}
404 \GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{-}{1}{-}}
405 \GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{-}{1}{-}}
```

\GTSoutIncidenceMonochromeRelation, \GTSoutIncMonoRel, \GTSoutIncMonRel, \GTSoIMR These macros will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 21.

```
406 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}
407 \GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}
408 \GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}
409 \GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}
```

\GTSnegatedOutIncidenceMonochrome, \GTSnegOutIncMono, \GTSnegOutIncMon, \GT-SnOIM These macros will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

See their [description](#) and usage examples on page 21.

```

410 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
411 \GTS@i{black}{black}{1}{1}}%
412 }
413 \GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}}
414 \GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}}
415 \GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}}

```

\GTSnegatedOutIncidenceMonochromeRelation, \GTSnegOutIncMonoRel, \GTSnegOutIncMonRel, \GTSnOIMR These macros will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode. See their [description](#) and usage examples on page 22.

```

416 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
417 \mathrel{\GTSnOIM/}%
418 }
419 \GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}
420 \GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}
421 \GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}

```

\GTSinIncidenceMonochrome, \GTSinIncMono, \GTSinIncMon, \GTSiIM These macros will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

See their [description](#) and usage examples on page 22.

```

422 \GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{}}{1}}
423 \GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{}}{1}}
424 \GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{}}{1}}
425 \GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{}}{1}}

```

\GTSinIncidenceMonochromeRelation, \GTSinIncMonoRel, \GTSinIncMonRel, \GTSiIMR These macros will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 23.

```

426 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
427 \GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
428 \GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
429 \GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}

```

\GTSnegatedInIncidenceMonochrome, \GTSnegInIncMono, \GTSnegInIncMon, \GTSnIIM These macros will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

See their [description](#) and usage examples on page 23.

```

430 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
431 \GTS@i{black}{black}{1}{1}}%
432 }
433 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{1}}
434 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{1}}
435 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{1}}

```

\GTSnegatedInIncidenceMonochromeRelation, \GTSnegInIncMonoRel, \GTSnegInIncMonRel, \GTSnIIMR These macros will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

See their [description](#) and usage examples on page 23.

```

436 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
437 \mathrel{\GTSnIIM/}%
438 }
439 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
440 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
441 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}

```

\GTSmetaNot, \GTSmN These macros will draw a meta-not or relation-not symbol. See their [description](#) and usage examples on page 24.

```

442 \GTS@newprotectedcommand{\GTSmN}{%
443 \text{\ensuremath{%
444 \raisebox{-0.44ex}{\scalebox{1}[1]{%
445 \raisebox{0ex}{\scalebox{1}[1]{\not$}}}%
446 \hspace{-1.46ex}%
447 \raisebox{1.2ex}{\scalebox{0.7}[0.7]{m}}}%
448 \hspace{0.2ex}%
449 }}%
450 }}%
451 }
452 \GTS@newcommand{\GTSmetaNot}{\GTSmN/}

```

\GTSmetaAnd, \GTSmA These macros will draw a meta-and or relations-and symbol. See their [description](#) and usage examples on page 24.

```

453 \GTS@newprotectedcommand{\GTSmA}{%
454 \text{\ensuremath{%
455 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
456 \raisebox{0ex}{\scalebox{1}[1]{\wedge$}}}%
457 \hspace{-1.46ex}%
458 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
459 }}%
460 }}%
461 }
462 \GTS@newcommand{\GTSmetaAnd}{\GTSmA/}

```

\GTSmetaAndRelation, \GTSmetaAndRel, \GTSmAR These macros will draw a meta-and symbol with relation spacing in math mode. See their [description](#) and usage examples on page 24.

```

463 \GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
464 \GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
465 \GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}

```

\GTSmetaOr, \GTSmO These macros will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have taken various meanings, whilst we want an unambiguous symbol. See their [description](#) and usage examples on page 24.

```

466 \GTS@newprotectedcommand{\GTSmO}{%
467 \text{\ensuremath{%
468 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
469 \raisebox{0ex}{\scalebox{1}[1]{\vee$}}}%
470 \hspace{-1.46ex}%
471 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
472 }}%
473 }}%
474 }
475 \GTS@newcommand{\GTSmetaOr}{\GTSmO/}

```

\GTSmetaOrRelation, \GTSmetaOrRel, \GTSmOR These macros will draw a meta-or symbol with relation spacing in math mode. See their [description](#) and usage examples on page 25.

```

476 \GTS@newcommand{\GTSmetaOrRelation}{\mathrel{\GTSmO/}}
477 \GTS@newcommand{\GTSmetaOrRel}{\mathrel{\GTSmO/}}
478 \GTS@newcommand{\GTSmOR}{\mathrel{\GTSmO/}}

```

\GTSxmetallogic, \GTSxml See their [description](#) on page 25.

```

479 \define@key{GTSxml}{not}[1]{\def\GTS@@drawMetaNot{1}}
480 \define@key{GTSxml}{n}[1]{\def\GTS@@drawMetaNot{1}}
481 \define@key{GTSxml}{and}[1]{\def\GTS@@drawMetaAnd{1}}

```

```

482 \define@key{GTSxml}{a}[1]{\def\GTS@@drawMetaAnd{1}}
483 \define@key{GTSxml}{or}[1]{\def\GTS@@drawMetaOr{1}}
484 \define@key{GTSxml}{o}[1]{\def\GTS@@drawMetaOr{1}}
485 \define@key{GTSxml}{relation}[1]{\def\GTS@@relation{1}}
486 \define@key{GTSxml}{rel}[1]{\def\GTS@@relation{1}}
487 \define@key{GTSxml}{r}[1]{\def\GTS@@relation{1}}
488
489 \newcommand{\GTSxml}[1] [] {%
490 \def\GTS@@drawMetaNot{%
491 \def\GTS@@drawMetaAnd{%
492 \def\GTS@@drawMetaOr{%
493 \def\GTS@@relation{%
494 \setkeys{GTSxml}{#1}%
495 \if\GTS@@drawMetaNot1%
496 \GTSmN/%
497 \else\if\GTS@@drawMetaAnd1%
498 \if\GTS@@relation{}1%
499 \GTSmAR/%
500 \else%
501 \GTSmA/%
502 \fi%
503 \else\if\GTS@@drawMetaOr1%
504 \if\GTS@@relation{}1%
505 \GTSmOR/%
506 \else%
507 \GTSmO/%
508 \fi%
509 \else%
510 Wrong call to \textbackslash{}GTSxml%
511 \fi%
512 \fi%
513 \fi%
514 }
515
516 \newcommand{\GTSxmetallogic}[1] [] {\GTSxml[#1]}

```

\GTSx See its [description](#) on page 25.

```

517 \newcommand{\GTSx}[2] [] {%
518 \if#2a%
519 \GTSxa[#1]%
520 \else\if#2i%
521 \GTSxi[#1]%
522 \else\if#2m%
523 \GTSxml[#1]%
524 \else%
525 Wrong call to \textbackslash{}GTSx%
526 \fi%
527 \fi%
528 \fi%
529 }

530 \endinput
531 \</package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 FAQ

Question: Why GraphTheorySymbol instead of GraphTheorySymbols? GraphTheorySymbol does contain many symbols.

Answer: Because I followed one common usage on CTAN.

Here is the list of all current packages with the word "symbol" in their name:

actuarialsymbol, adfsymbols, compact-symbols, cookingsymbols, FdSymbol, jpnedumathsymbols, KKsymbols, listofsymbols, losymbol, maths-symbols, mdsymbol, MnSymbol, open-everyday-symbols, qsymbols, SVRsymbols, symbolindex, The Comprehensive LaTeX Symbol List, tikz-cookingsymbols, tikzsymbols.

7 "symbol" plus GraphTheorySymbol 8, 12 "symbols".

"symb" abbreviation, "sym", "sy" and "s" are also quite common;

oPlotSymb did it its own way ;) XD.

Because of the goal of GraphTheorySymbol, I was mainly looking at packages like FdSymbol, mdsymbol, MnSymbol, and The Comprehensive LaTeX Symbol List before creating my package; hence I followed this apparent majority, instead of listening to my thought that Symbols would be better.

After the package was uploaded to CTAN for a version update, this question came again, but CTAN admins were reluctant to change the package name afterward.

Otherwise, it is a matter of hours to modify and upload GraphTheorySymbols with version 2.0.0 to CTAN.

6 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

7 Change History

v0.1.0 – 2026-08-01		avoid too big left margins, and use	
General: Initial version	1	paragraph command instead	1
v1.0.0 – 2026-08-26		v1.5.2 – 2026-09-14	
General: First public release	1	General: Add A4, USLetter, and Book variants	
v1.1.0 – 2026-08-29		of PDFs.	1
General: Added a memo	1	v1.6.0 – 2026-09-18	
v1.1.1 – 2026-08-30		General: Add verbatim code for compound	
General: Small corrections and improvements	1	adjacency in memo.	1
v1.2.0 – 2026-09-06		v1.6.1 – 2026-09-19	
General: CTAN URL, new macros for same set		General: Rename test(.tex, etc.) to	
of symbols	1	graph-theory-symbol-test(.tex, etc.).	1
v1.2.1 – 2026-09-08		v1.7.0 – 2026-09-21	
General: Smaller tests output	1	General: Add FAQ section. Minors spacing and	
v1.3.0 – 2026-09-08		newpage corrections. Put table of contents	
General: New macros for monochrome symbols	1	and Introduction on their own page. Use	
v1.4.0 – 2026-09-12		tcolorbox for more visible examples.	
General: New macros for symbols for logical		Disclaimer for documentation of macros	
operations on relations	1	with an at sign.	1
v1.5.0 – 2026-09-12		v1.8.0 – 2026-09-28	
General: One macro to rule them all	1	General: Forward and backward hyperlinks	
v1.5.1 – 2026-09-13		between macros descriptions and	
General: Use noprint option of doc package to		implementations.	1

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